

GLMMs and Generalized Estimating Equations

Part VII replaced independent errors by a covariance matrix with structure; Part VIII has so far replaced the normal response by an exponential dispersion family. This chapter does both at once, and the combination is not the sum of its parts. In a linear model the mean is unaffected by what happens to the covariance, so a random effect changes the standard errors and leaves the coefficients alone. Once the link is nonlinear that is false: averaging a curve over a random intercept does not give the same curve back, so the model has *two* sets of coefficients — one for a single cluster, one for the population — and they are not equal.

Two families of methods follow from that fork. If the interest is in what happens within a cluster, and a full probability model is acceptable, the answer is the generalized linear mixed model: random effects inside the linear predictor and a likelihood obtained by integrating them out, numerically, with different approximations giving visibly different answers. If the interest is in the population average, and no model for the dependence is wanted, the answer is generalized estimating equations: solve a quasi-score built from a *working* correlation that is allowed to be wrong, and price the estimate with a sandwich. The second route buys robustness by giving up the likelihood, and with it the deviance, the random-effect predictions, and any tolerance for data that go missing for the wrong reasons.

What you need. [Chapter 34](#) throughout (the generalized linear model Definition 34.2.1 and Definition 34.2.2, the likelihood equations Theorem 34.3.1, iteratively reweighted least squares Theorem 34.4.1, the asymptotics Theorem 34.5.1), [Chapter 35](#) (the logistic model Definition 35.1.1, non-collapsibility Example (The odds ratio does not collapse) and case-control sampling Theorem 35.5.1), [Chapter 36](#) and [Chapter 37](#) (Poisson regression Definition 37.1.1 and the negative binomial Definition 37.4.1). From Part VII: the linear mixed model Definition 32.2.1, best linear unbiased prediction Theorem 32.3.1 and its shrinkage reading, REML Definition 32.4.1, the boundary problem Theorem 32.5.2, the covariance models Definition 33.4.1 and the working-covariance result Proposition 33.4.1. From Part V: the sandwich covariance Theorem 21.4.2. [Chapter 29](#) (AIC and BIC, Definition 29.2.2) and [Chapter 25](#) (what a coefficient means) are used in places. [Chapter 38](#) supplies the quasi-likelihood machinery this chapter generalizes, while the missingness conditions that [Section 40.4](#) needs are stated in [Chapter 41](#), which comes after this one. [Section 40.1](#) uses the moment generating function of the multivariate normal from [Chapter 3](#) (Theorem 3.2.1) to compute what a random intercept does to a marginal mean.

40.-1.1 Roadmap

- [Marginal versus conditional models](#)
- [Generalized linear mixed models](#)
- [Integrated likelihood approximations](#)
- [Generalized estimating equations](#)
- [Working correlation](#)
- [Summary and notes](#)